

RTAB-Map Navigation

1. Content Description

This section explains how to implement RTAB-Map navigation by combining a vehicle chassis, LiDAR, depth camera, and Navigation2.

This section requires entering commands in the terminal. The appropriate terminal depends on your motherboard type. This lesson uses the Raspberry Pi 5 as an example. For Raspberry Pi and Jetson Nano boards, you need to open a terminal on the host computer and enter the command to enter the Docker container. Once inside the Docker container, enter the commands mentioned in this lesson in the terminal. For instructions on entering the Docker container from the host computer, refer to **[01. Robot Configuration and Operation Guide] -- [5.Enter Docker (For JETSON Nano and RPi 5)]**.

For Orin and RDK boards, simply open the terminal and enter the commands mentioned in this lesson.

2. Preparation

Due to performance limitations, Raspberry Pi 5 and Jetson Nano cannot smoothly run the RTAB-Map algorithm within the motherboard's Docker container. Therefore, a virtual machine is needed to assist in implementation. For the vehicle and the virtual machine to achieve distributed communication, the following two steps are required:

- They must be on the same local area network (LAN). The simplest way to achieve this is to connect to the same Wi-Fi network.
- Their ROS_DOMAIN_IDs must be consistent. The default ROS_DOMAIN_ID for both the vehicle and the virtual machine is 30. If they are different, the virtual machine's ROS_DOMAIN_ID needs to be modified. Modify the `~/.bashrc` file, change the ROS_DOMAIN_ID value to match the vehicle's, save and exit, then enter the command `source ~/.bashrc` to refresh the environment variables.
- Check if distributed communication is achieved. Run the driver program on the motherboard, then enter `ros2 node list` on the virtual machine. If the underlying data lists, communication between the two is successful.

Orin motherboards can run directly on the motherboard, and if RDK doesn't run smoothly, it can be run on a virtual machine for visualization.

3. Program Startup

For PI5/JETSON NANO controllers, you need to enter a Docker container first; for RDK X5 and Orin boards, this is not required.

Entering a Docker container (see the Docker course for steps: 4. Docker Startup Script).

All commands below must be executed within the same Docker container (see the Docker course for steps: 3. Docker Commit and Multi-Terminal Entry).

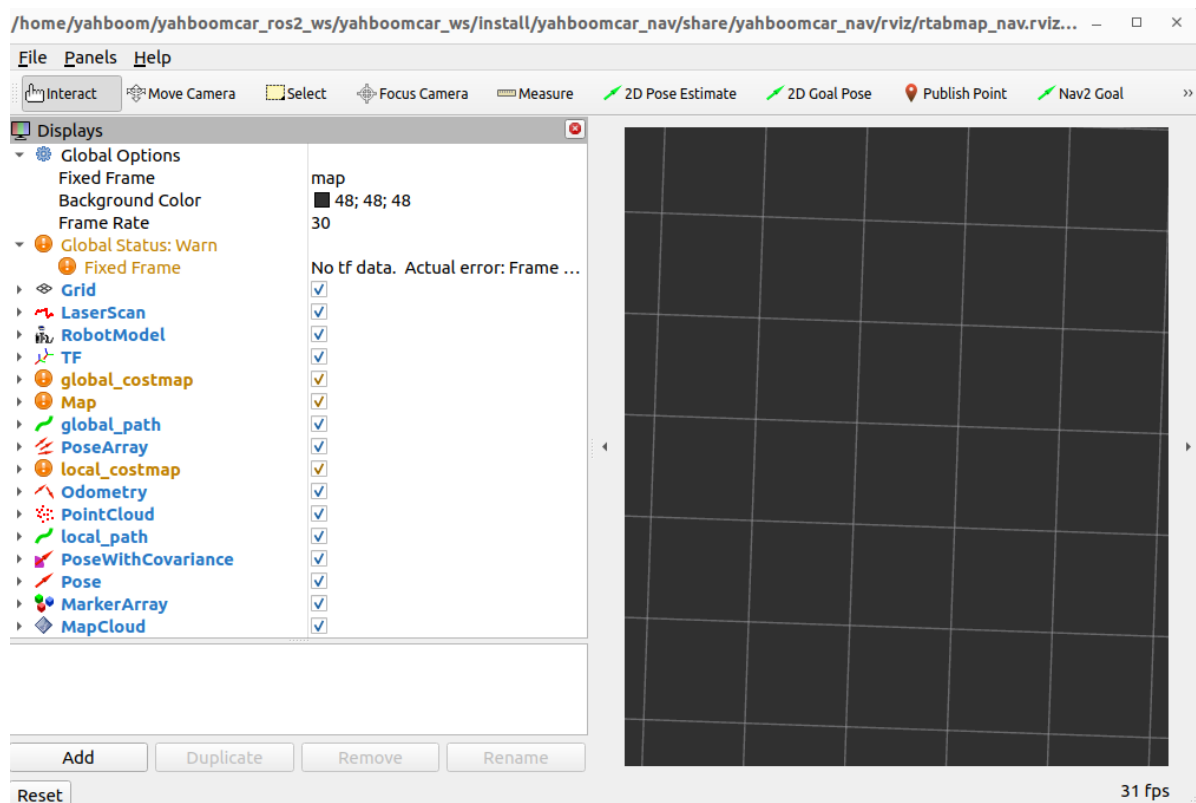
First, enter the following command in the vehicle terminal to start the depth camera.

```
ros2 launch ascamera hp60c.launch.py
```

```
root@raspberrypi: ~
File Edit Tabs Help
CE17000849 ]'s parameter:
[ascamera_node-1] [INFO] [1755082632.031565034] [ascamera_hp60c.camera_publisher]: irfx: 425
[ascamera_node-1] [INFO] [1755082632.031606847] [ascamera_hp60c.camera_publisher]: irfy: 425
[ascamera_node-1] [INFO] [1755082632.031626199] [ascamera_hp60c.camera_publisher]: ircx: 314.
577
[ascamera_node-1] [INFO] [1755082632.031646236] [ascamera_hp60c.camera_publisher]: ircy: 237.
151
[ascamera_node-1] [INFO] [1755082632.031660624] [ascamera_hp60c.camera_publisher]: rgbfx: 571
[ascamera_node-1] [INFO] [1755082632.031674735] [ascamera_hp60c.camera_publisher]: rgbfy: 571
[ascamera_node-1] [INFO] [1755082632.031689179] [ascamera_hp60c.camera_publisher]: rgbcx: 332
.029
[ascamera_node-1] [INFO] [1755082632.031704142] [ascamera_hp60c.camera_publisher]: rgbcy: 235
.042
[ascamera_node-1]
[ascamera_node-1] [INFO] [1755082632.034756871] [ascamera_hp60c.camera_publisher]: publish de
pth info
[ascamera_node-1] [INFO] [1755082632.036174044] [ascamera_hp60c.camera_publisher]: publish co
lor(rgb) info
[ascamera_node-1] [INFO] [1755082632.038389828] [ascamera_hp60c.camera_publisher]: publish tf
info
[ascamera_node-1] [INFO] [1755082756.310351452] [ascamera_hp60c.camera_publisher]: 2025-08-13
18:59:16[INFO] [CameraHp60c.cpp] [278] [stopStreaming] stop streaming
```

Enter the following command in the virtual machine terminal for rviz visualization.

```
ros2 launch yahboomcar_nav display_rtabmap_nav_launch.py
```



Wait for RTAB navigation to start.

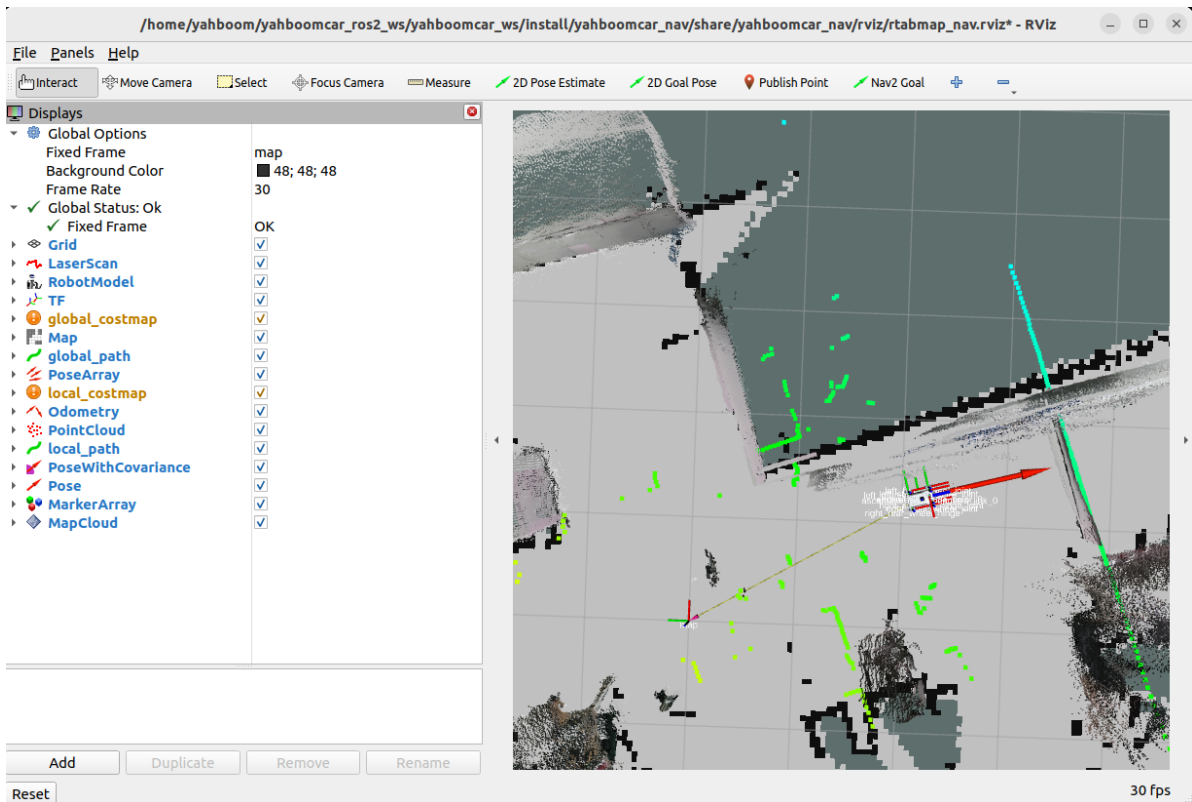
Enter the following command in the terminal to start the chassis and camera.

```
ros2 launch yahboomcar_nav navigation_rtabmap_launch.py
```

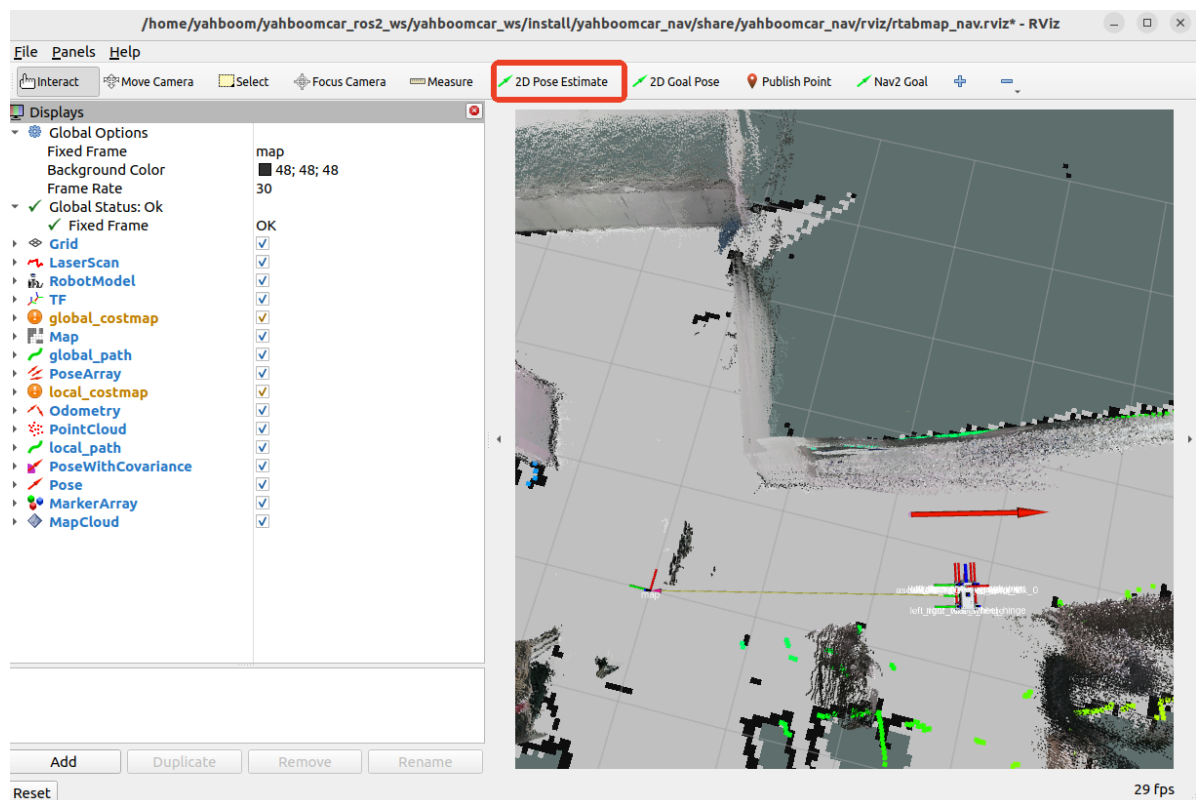
```
root@raspberrypi: ~/ros
File Edit Tabs Help

[rtabmap-14] [INFO] [1755084375.504523952] [rtabmap]: rtabmap: subscribe_scan = true
[rtabmap-14] [INFO] [1755084375.504535044] [rtabmap]: rtabmap: subscribe_scan_cloud = false
[rtabmap-14] [INFO] [1755084375.504547544] [rtabmap]: rtabmap: subscribe_scan_descriptor = false
[rtabmap-14] [INFO] [1755084375.504560859] [rtabmap]: rtabmap: topic_queue_size = 10
[rtabmap-14] [INFO] [1755084375.504573488] [rtabmap]: rtabmap: sync_queue_size = 10
[rtabmap-14] [INFO] [1755084375.504823005] [rtabmap]: rtabmap: qos_image = 0
[rtabmap-14] [INFO] [1755084375.504839430] [rtabmap]: rtabmap: qos_camera_info = 0
[rtabmap-14] [INFO] [1755084375.504850597] [rtabmap]: rtabmap: qos_scan = 0
[rtabmap-14] [INFO] [1755084375.504864523] [rtabmap]: rtabmap: qos_odom = 0
[rtabmap-14] [INFO] [1755084375.504876578] [rtabmap]: rtabmap: qos_user_data = 0
[rtabmap-14] [INFO] [1755084375.504890597] [rtabmap]: rtabmap: approx_sync = true
[rtabmap-14] [INFO] [1755084375.504920356] [rtabmap]: Setup rgbd callback
[rtabmap-14] [INFO] [1755084375.511538409] [rtabmap]:
[rtabmap-14] rtabmap subscribed to (approx sync):
[rtabmap-14] /rgbd_image \
[rtabmap-14] /scan
[rtabmap-14] [INFO] [1755084377.341748082] [rtabmap]: rtabmap (455): Rate=1.00s, Limit=0.000s, Conversion=0.0009s, RTAB-Map=0.1649s, Map
ps update=0.0003s pub=0.0002s delay=0.3666s (local map=201, WM=201)
[rtabmap-14] [INFO] [1755084378.972419639] [rtabmap]: rtabmap (456): Rate=1.00s, Limit=0.000s, Conversion=0.0048s, RTAB-Map=0.1688s, Ma
ps update=0.2560s pub=0.0810s delay=0.9023s (local map=201, WM=201)
[rtabmap-14] [INFO] [1755084379.579533737] [rtabmap]: rtabmap (457): Rate=1.00s, Limit=0.000s, Conversion=0.0008s, RTAB-Map=0.1201s, Ma
ps update=0.0013s pub=0.0910s delay=0.3137s (local map=201, WM=201)
[rtabmap-14] [INFO] [1755084380.840602559] [rtabmap]: rtabmap (458): Rate=1.00s, Limit=0.000s, Conversion=0.0007s, RTAB-Map=0.1268s, Ma
ps update=0.0010s pub=0.0506s delay=0.3801s (local map=201, WM=201)
[rtabmap-14] [INFO] [1755084382.370282434] [rtabmap]: rtabmap (459): Rate=1.00s, Limit=0.000s, Conversion=0.0009s, RTAB-Map=0.1719s, Ma
ps update=0.0007s pub=0.4397s delay=0.8147s (local map=201, WM=201)
[ekf_node-6] Failed to meet update rate! Took 0.04897614399999999235seconds. Try decreasing the rate, limiting sensor output frequency
, or limiting the number of sensors.
[rtabmap-14] [INFO] [1755084383.530832289] [rtabmap]: rtabmap (460): Rate=1.00s, Limit=0.000s, Conversion=0.0005s, RTAB-Map=0.1173s, Ma
ps update=0.0008s pub=0.2621s delay=0.4818s (local map=201, WM=201)
```

At the same time, rviz will display the map.



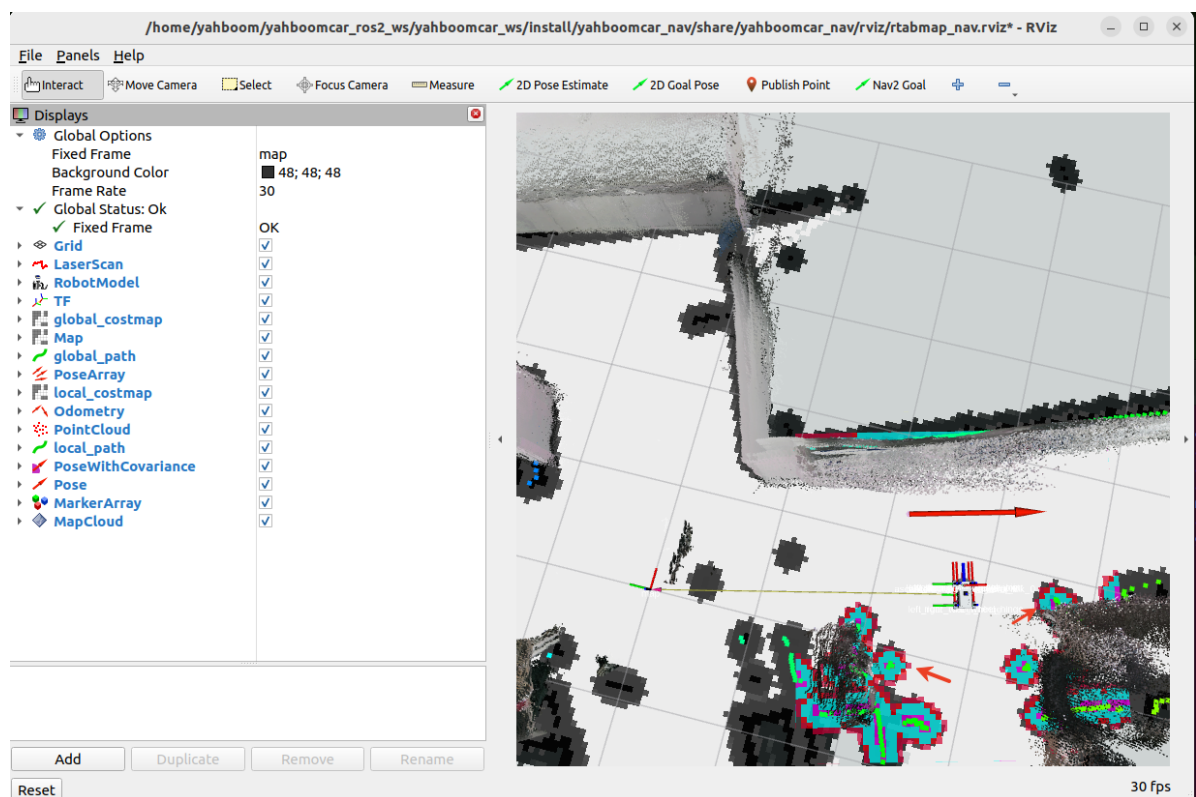
- From rviz, you can see that the robot automatically estimates its initial position, eliminating the need for manual position initialization.
- If the car's initial position is significantly off, use the [2D Pose Estimate] tool in the RVIZ toolbar to provide an approximate position for quick positioning.



Enter the following command in the terminal to start navigation2:

```
ros2 launch yahboomcar_nav rtabmap_nav_launch.py
```

After everything is successfully launched, an expansion area will appear, as shown below.



Then use the [Nav2 Goal] tool in RVIZ to specify a target point for the car, and it will navigate to that point.


```

root@raspberrypi1:~# ros2 node info /rgbd_sync
/rgbd_sync
Subscribers:
  /ascamera_hp60c/camera_publisher/depth0/image_raw: sensor_msgs/msg/Image
  /ascamera_hp60c/camera_publisher/rgb0/camera_info: sensor_msgs/msg/CameraInfo
  /ascamera_hp60c/camera_publisher/rgb0/image: sensor_msgs/msg/Image
  /parameter_events: rcl_interfaces/msg/ParameterEvent
Publishers:
  /diagnostics: diagnostic_msgs/msg/DiagnosticArray
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rgbd_image: rtabmap_msgs/msg/RGBDImage
  /rgbd_image/compressed: rtabmap_msgs/msg/RGBDImage
  /rosout: rcl_interfaces/msg/Log
Service Servers:
  /rgbd_sync/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /rgbd_sync/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /rgbd_sync/get_parameters: rcl_interfaces/srv/GetParameters
  /rgbd_sync/list_parameters: rcl_interfaces/srv/ListParameters
  /rgbd_sync/set_parameters: rcl_interfaces/srv/SetParameters
  /rgbd_sync/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
Service Clients:

Action Servers:

Action Clients:

root@raspberrypi1:~#

```

View the /rtabmap node

```
ros2 node info /rtabmap
```

```

/rtabmap
Subscribers:
  /global_pose: geometry_msgs/msg/PoseWithCovarianceStamped
  /goal: geometry_msgs/msg/PoseStamped
  /goal_node: rtabmap_msgs/msg/Goal
  /gps/fix: sensor_msgs/msg/NavSatFix
  /imu: sensor_msgs/msg/Imu
  /initialpose: geometry_msgs/msg/PoseWithCovarianceStamped
  /landmark_detection: rtabmap_msgs/msg/LandmarkDetection
  /landmark_detections: rtabmap_msgs/msg/LandmarkDetections
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rgbd_image: rtabmap_msgs/msg/RGBDImage
  /rtabmap/republish_node_data: std_msgs/msg/Int32MultiArray
  /scan: sensor_msgs/msg/LaserScan
  /user_data_async: rtabmap_msgs/msg/UserData
Publishers:
  /cloud_ground: sensor_msgs/msg/PointCloud2
  /cloud_map: sensor_msgs/msg/PointCloud2
  /cloud_obstacles: sensor_msgs/msg/PointCloud2
  /diagnostics: diagnostic_msgs/msg/DiagnosticArray
  /global_path: nav_msgs/msg/Path
  /global_path_nodes: rtabmap_msgs/msg/Path
  /goal_out: geometry_msgs/msg/PoseStamped
  /goal_reached: std_msgs/msg/Bool
  /grid_prob_map: nav_msgs/msg/OccupancyGrid
  /info: rtabmap_msgs/msg/Info
  /labels: visualization_msgs/msg/MarkerArray
  /landmarks: geometry_msgs/msg/PoseArray
  /local_grid_empty: sensor_msgs/msg/PointCloud2
  /local_grid_ground: sensor_msgs/msg/PointCloud2
  /local_grid_obstacle: sensor_msgs/msg/PointCloud2
  /local_path: nav_msgs/msg/Path
  /local_path_nodes: rtabmap_msgs/msg/Path
  /localization_pose: geometry_msgs/msg/PoseWithCovarianceStamped
  /map: nav_msgs/msg/OccupancyGrid
  /mapData: rtabmap_msgs/msg/MapData
  /mapGraph: rtabmap_msgs/msg/MapGraph

```

/mapOdomCache: rtabmap_msgs/msg/MapGraph
/mapPath: nav_msgs/msg/Path
/octomap_binary: octomap_msgs/msg/Octomap
/octomap_empty_space: sensor_msgs/msg/PointCloud2
/octomap_full: octomap_msgs/msg/Octomap
/octomap_global_frontier_space: sensor_msgs/msg/PointCloud2
/octomap_grid: nav_msgs/msg/OccupancyGrid
/octomap_ground: sensor_msgs/msg/PointCloud2
/octomap_obstacles: sensor_msgs/msg/PointCloud2
/octomap_occupied_space: sensor_msgs/msg/PointCloud2
/parameter_events: rcl_interfaces/msg/ParameterEvent
/rosout: rcl_interfaces/msg/Log
/tf: tf2_msgs/msg/TFMessage

Service Servers:

/rtabmap/add_link: rtabmap_msgs/srv/AddLink
/rtabmap/backup: std_srvs/srv/Empty
/rtabmap/cancel_goal: std_srvs/srv/Empty
/rtabmap/cleanup_local_grids: rtabmap_msgs/srv/CleanupLocalGrids
/rtabmap/describe_parameters: rcl_interfaces/srv/DescribeParameters
/rtabmap/detect_more_loop_closures: rtabmap_msgs/srv/DetectMoreLoopClosures
/rtabmap/get_map: nav_msgs/srv/GetMap
/rtabmap/get_map_data: rtabmap_msgs/srv/GetMap
/rtabmap/get_map_data2: rtabmap_msgs/srv/GetMap2
/rtabmap/get_node_data: rtabmap_msgs/srv/GetNodeData
/rtabmap/get_nodes_in_radius: rtabmap_msgs/srv/GetNodesInRadius
/rtabmap/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
/rtabmap/get_parameters: rcl_interfaces/srv/GetParameters
/rtabmap/get_plan: nav_msgs/srv/GetPlan
/rtabmap/get_plan_nodes: rtabmap_msgs/srv/GetPlan
/rtabmap/get_prob_map: nav_msgs/srv/GetMap
/rtabmap/global_bundle_adjustment: rtabmap_msgs/srv/GlobalBundleAdjustment
/rtabmap/list_labels: rtabmap_msgs/srv/ListLabels
/rtabmap/list_parameters: rcl_interfaces/srv/ListParameters
/rtabmap/load_database: rtabmap_msgs/srv/LoadDatabase
/rtabmap/log_debug: std_srvs/srv/Empty
/rtabmap/log_error: std_srvs/srv/Empty
/rtabmap/log_info: std_srvs/srv/Empty
/rtabmap/log_warning: std_srvs/srv/Empty
/rtabmap/octomap_binary: octomap_msgs/srv/GetOctomap
/rtabmap/octomap_full: octomap_msgs/srv/GetOctomap
/rtabmap/pause: std_srvs/srv/Empty
/rtabmap/publish_map: rtabmap_msgs/srv/PublishMap
/rtabmap/remove_label: rtabmap_msgs/srv/RemoveLabel
/rtabmap/reset: std_srvs/srv/Empty
/rtabmap/resume: std_srvs/srv/Empty
/rtabmap/set_goal: rtabmap_msgs/srv/SetGoal
/rtabmap/set_label: rtabmap_msgs/srv/SetLabel
/rtabmap/set_mode_localization: std_srvs/srv/Empty
/rtabmap/set_mode_mapping: std_srvs/srv/Empty
/rtabmap/set_parameters: rcl_interfaces/srv/SetParameters
/rtabmap/set_parameters_atomically:

rcl_interfaces/srv/SetParametersAtomically

/rtabmap/trigger_new_map: std_srvs/srv/Empty
/rtabmap/update_parameters: std_srvs/srv/Empty

Service Clients:

/rtabmap/describe_parameters: rcl_interfaces/srv/DescribeParameters
/rtabmap/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
/rtabmap/get_parameters: rcl_interfaces/srv/GetParameters

```
/rtabmap/list_parameters: rcl_interfaces/srv/ListParameters
/rtabmap/set_parameters: rcl_interfaces/srv/SetParameters
/rtabmap/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
Action Servers:

Action Clients:
```